

Sébastien Roux

Mechanical Engineer, Robotics & Hardware Design

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Professional Summary

Mechanical engineer who likes building systems that survive the real world, not just the lab bench. Graduating from EPFL (Master in Design and Production) with hands-on experience across four robotics and energy R&D programs, owning hardware from concept and CAD through fabrication, bench testing, and in-situ field validation.

Professional Experience

ODEWA, R&D Engineering Intern

September 2025 – February 2026

- Owned the full product lifecycle for an automated underwater waste-extraction machine deployed in French Polynesia: wrote specifications, designed the system in CATIA, and built and tested a working prototype in-situ.
- Applied structural finite element analysis in Abaqus to size key components against local fabrication and material constraints, exhibiting cradle-to-grave ownership across design, build, and field validation.
- Prototype removed and cleaned 1 tonne of waste in 2 days during field testing; findings are informing a second-generation prototype currently in development.

AIRARO SAS, R&D Engineering Intern

March 2026 – September 2026

- Analyzed live operating data from a SWAC energy plant and built a sizing tool in Python and MATLAB combining thermal and hydraulic modeling to evaluate design tradeoffs.
- Design changes identified through this analysis improved system COP by 25 percent and were integrated directly into the company's business plan.

Robotics and Engineering Projects

3-DOF Robotic Platform, Reconfigurable Robotics Lab (EPFL)

September 2024 – February 2025

- Redesigned the structure of an existing 3-DOF robotic platform in SolidWorks, running FEA to support human body loads, and laser-cut the metal components for fabrication.
- Built a Python and MATLAB data pipeline to extract periodic locomotion from multimodal video and sensor inputs, then reproduced it on the physical platform and validated it with motion capture testing.

Biomimetic Foot Structures, BioRob Lab (EPFL)

February 2025 – July 2025

- Designed compliant biomimetic feet for a quadruped robot, fabricated with flexible 3D printing, and built a dedicated test bench to characterize their mechanical response through rapid iteration.
- Integrated the design on the robot and validated it with motion capture testing, achieving a 15 percent gain in locomotion efficiency.

In-Wheel Electric Propulsion, EPFL Racing Team (Formula Student)

February 2022 – August 2023

- Designed and optimized an in-wheel gearbox and propulsion system in the 3DEXPERIENCE (CATIA) environment under tight thermal, mass, and regulatory constraints, partnering with a cross-functional engineering team to take the design from concept to a raced vehicle.

Energy Storage Optimization, collaboration with ZF Group

2024

- Designed a reduced-scale physical demonstrator with Arduino-integrated hardware and an energy-management algorithm for a battery-swap station serving electric commercial vehicles (eTrucks, eBuses, eTrailers), complemented by a business model study.

Education

Master in Mechanical Engineering, EPFL

2026

Specialization in Design and Production; minor in Management and Technology of Entrepreneurship

Bachelor in Mechanical Engineering, EPFL

2023

French Baccalauréat, Lycée Français de Toronto

2019

Languages

French (native, C2), English (fluent, C2), Spanish (basic, A2)